

# A reference-cohort summary estimator for genetic and gene–environment variance components

SUMMIT technical report

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## Abstract

GENIE estimates additive genetic (G), gene–environment (G×E), noise–environment (N×E), and residual variance components from one individual-level cohort. This report first writes that estimator as one matrix normal equation. Its trait-dependent right-hand side is exactly a function of marginal GWAS and GWIS cross-products, but that observation alone does not make the kernel Gram matrix reusable across samples. We derive the additional reference moments needed when G and G×E LD quantities are estimated in every individual with a measured environment, while each trait uses its own, potentially smaller cohort. The essential step is to transfer the same-individual and different-individual pieces of every genetic kernel-product trace separately. The G×E same-individual piece is a non-universal fourth moment; SUMMIT estimates it during the randomized, in-memory reference pass with an order-two probe U-statistic and  $O(N_R K)$  memory independent of the number of probes. N×E entries are computed exactly in each trait cohort. We state the estimand and scaling conventions, the sufficient summary-data contract, the limitations created by variant-specific missingness, and an audit of covariate handling in the released GENIE implementation.

## 1 Estimator roadmap

The construction is easiest to follow as five operations.

1. In each cohort, standardize the continuous environment, form one fixed-effect projector, and use it consistently for phenotype, genotype, and G×E features.
2. Express GENIE as the normal equation  $\mathbf{T}\boldsymbol{\theta} = \mathbf{q}$  for the four kernel families G, G×E, N×E, and residual.
3. Obtain the genetic entries of  $\mathbf{q}$  from trait-specific marginal GWAS/GWIS cross-products and obtain the genetic block of  $\mathbf{T}$  from reference LD summaries.
4. Convert the reference genetic block from  $N_R$  to the trait sample size  $N_S$  by transferring its same-person and different-person pieces separately. A single squared sample-size multiplier is generally wrong.
5. Compute the inexpensive N×E entries in the trait cohort, solve the assembled normal equation, and jackknife over variant blocks.

The sections below derive each operation. We first show one annotation so the matrix is visible, then give the general  $K$ -annotation notation used by the software.

## 2 Step 1: cohort transformation and generative model

We follow the notation of Kerin and Marchini [1]. Cohort  $S$  is the trait study and cohort  $R$  is the reusable reference. They share an ordered set of  $M$  variants and the same annotation weights  $A_{jk} \geq 0$ ,  $k = 1, \dots, K$ , with  $M_k = \sum_j A_{jk}$ . Their individuals need not be the same. Let  $\mathbf{C}_S$  contain an intercept, the environment, and the other fixed effects used by the trait analysis. Define

$$\mathbf{P}_S = \mathbf{I} - \mathbf{C}_S(\mathbf{C}_S^\top \mathbf{C}_S)^+ \mathbf{C}_S^\top, \quad r_S = \text{tr}(\mathbf{P}_S) = N_S - \text{rank}(\mathbf{C}_S), \quad (1)$$

where  $(\cdot)^+$  is the Moore–Penrose inverse. The same definitions apply to  $R$ .

For a continuous environment, SUMMIT first transforms the observed values in each retained cohort to

$$e_i = \frac{E_i - \bar{E}}{s_E}, \quad s_E^2 = \frac{1}{N - d_E} \sum_i (E_i - \bar{E})^2, \quad d_E \in \{0, 1\}. \quad (2)$$

The default is  $d_E = 1$ , so the effect units are per environmental standard deviation. The standardized  $\mathbf{e}$  is then included in  $\mathbf{C}$  as a fixed main effect. This transformation is performed separately in  $R$  and  $S$  because the reference and trait masks can differ; the manifest records the mean, standard deviation, degrees-of-freedom convention, and design digest. Binary environments can be analyzed with the same algebra, although Eq. (2) changes their numerical contrast and can make  $N \times E$  inseparable from the residual component in special cases discussed below.

Let  $\tilde{\mathbf{y}}_S$  denote the phenotype before fixed-effect adjustment. SUMMIT uses

$$\mathbf{y}_S = \mathbf{P}_S \tilde{\mathbf{y}}_S \sqrt{\frac{r_S}{\tilde{\mathbf{y}}_S^\top \mathbf{P}_S \tilde{\mathbf{y}}_S}}, \quad \mathbf{y}_S^\top \mathbf{y}_S = r_S. \quad (3)$$

The primary variance proportions therefore describe variation remaining after the declared fixed effects. Multiplying them by the recorded post-projection phenotype-variance fraction instead expresses their contribution relative to the preprojection phenotype variance; that is a different denominator, not a second estimator.

Let  $\mathbf{G}_S$  be a centered, pre-scaled genotype matrix and let  $\mathbf{e}_S$  be a centered environment measured in standard-deviation units. The default one-environment GENIE generative model, before fixed-effect projection, is

$$\tilde{\mathbf{y}}_S = \mathbf{G}_S \tilde{\boldsymbol{\beta}} + \text{diag}(\mathbf{e}_S) \mathbf{G}_S \tilde{\boldsymbol{\alpha}} + \text{diag}(\mathbf{e}_S) \boldsymbol{\delta} + \mathbf{C}_S \boldsymbol{\gamma} + \boldsymbol{\varepsilon}. \quad (4)$$

The published framework allows several environments; the present derivation applies to one environment at a time. GENIE uses the naturally scaled projected columns  $\mathbf{P}_S \mathbf{G}_S$  and  $\mathbf{P}_S \text{diag}(\mathbf{e}_S) \mathbf{G}_S$ . SUMMIT’s default instead restandardizes those columns after projection:

$$\begin{aligned} \mathbf{U}_S &= \mathbf{P}_S \mathbf{G}_S, & \mathbf{X}_{S,::j} &= \mathbf{U}_{S,::j} \sqrt{\frac{r_S}{\mathbf{U}_{S,::j}^\top \mathbf{U}_{S,::j}}}, \\ \mathbf{V}_S &= \mathbf{P}_S \text{diag}(\mathbf{e}_S) \mathbf{G}_S, & \mathbf{W}_{S,::j} &= \mathbf{V}_{S,::j} \sqrt{\frac{r_S}{\mathbf{V}_{S,::j}^\top \mathbf{V}_{S,::j}}}. \end{aligned} \quad (5)$$

Thus every retained column of  $\mathbf{X}_S$  and  $\mathbf{W}_S$  has squared norm  $r_S$ . This is the  $G \times E$  analogue of SUMMIT’s covariate-residualized, restandardized additive feature definition [3]. The two choices define different random-effect scalings, but the method-of-moments algebra below is the same when

all traces and scores use one choice coherently. The external implementation uses the standardized choice. Define

$$\begin{aligned} \mathbf{K}_{Sk}^G &= \frac{\mathbf{X}_S \text{diag}(\mathbf{A}_{\cdot k}) \mathbf{X}_S^\top}{M_k}, & \mathbf{K}_{Sk}^I &= \frac{\mathbf{W}_S \text{diag}(\mathbf{A}_{\cdot k}) \mathbf{W}_S^\top}{M_k}, \\ \mathbf{D}_S &= \mathbf{P}_S \text{diag}(\mathbf{e}_S^2) \mathbf{P}_S, & \mathbf{K}_S^0 &= \mathbf{P}_S. \end{aligned} \quad (6)$$

The superscript  $I$  denotes interaction and avoids confusing an environment vector with an expectation.

The residualized phenotype is generated under the second-moment model

$$\mathbf{y}_S = \sum_{k=1}^K \mathbf{X}_S \beta_k + \sum_{k=1}^K \mathbf{W}_S \alpha_k + \mathbf{P}_S \text{diag}(\mathbf{e}_S) \boldsymbol{\delta} + \mathbf{P}_S \boldsymbol{\varepsilon}, \quad (7)$$

with mutually uncorrelated, mean-zero effects

$$\begin{aligned} \text{Cov}(\beta_k) &= \frac{\sigma_{Gk}^2}{M_k} \text{diag}(\mathbf{A}_{\cdot k}), & \text{Cov}(\alpha_k) &= \frac{\sigma_{Ik}^2}{M_k} \text{diag}(\mathbf{A}_{\cdot k}), \\ \text{Cov}(\boldsymbol{\delta}) &= \sigma_D^2 \mathbf{I}, & \text{Cov}(\boldsymbol{\varepsilon}) &= \sigma_0^2 \mathbf{I}. \end{aligned}$$

For overlapping annotations this is understood as a linear covariance model; the coefficients need not each be nonnegative. Conditional on the design,

$$\boldsymbol{\Sigma}_S = \text{Cov}(\mathbf{y}_S \mid \mathbf{X}_S, \mathbf{W}_S, \mathbf{e}_S, \mathbf{C}_S) = \sum_k \sigma_{Gk}^2 \mathbf{K}_{Sk}^G + \sum_k \sigma_{Ik}^2 \mathbf{K}_{Sk}^I + \sigma_D^2 \mathbf{D}_S + \sigma_0^2 \mathbf{P}_S. \quad (8)$$

No Gaussian assumption on the effects is required for the moment equations.

### 3 Step 2: GENIE as one matrix normal equation

Index the  $2K + 2$  kernels in Eq. (8) by  $\mathcal{K}_S = (\mathbf{K}_{S1}^G, \dots, \mathbf{K}_{SK}^G, \mathbf{K}_{S1}^I, \dots, \mathbf{K}_{SK}^I, \mathbf{D}_S, \mathbf{P}_S)$  and collect their coefficients in  $\boldsymbol{\theta}$ . GENIE minimizes the Frobenius loss

$$\left\| \mathbf{y}_S \mathbf{y}_S^\top - \sum_a \theta_a \mathcal{K}_{Sa} \right\|_F^2. \quad (9)$$

Differentiation gives the normal equations

$$\mathbf{T}_S \hat{\boldsymbol{\theta}} = \mathbf{q}_S, \quad (T_S)_{ab} = \text{tr}(\mathcal{K}_{Sa} \mathcal{K}_{Sb}), \quad (q_S)_a = \mathbf{y}_S^\top \mathcal{K}_{Sa} \mathbf{y}_S. \quad (10)$$

This is precisely Supplemental Note S1 of GENIE after writing the projected kernels explicitly [2]. GENIE estimates large traces by Hutchinson random projection: for independent isotropic probes  $\mathbf{z}_v$ ,

$$\text{tr}(\mathcal{K}_a \mathcal{K}_b) = \mathbb{E}_v[(\mathcal{K}_a \mathbf{z}_v)^\top (\mathcal{K}_b \mathbf{z}_v)]. \quad (11)$$

For one annotation, define  $G = \mathbf{K}^G$ ,  $I = \mathbf{K}^I$ ,  $D = \mathbf{D}$ , and  $0 = \mathbf{P}$ , and write  $T_{ab} = \text{tr}(\mathbf{K}^a \mathbf{K}^b)$  and  $t_a = \text{tr}(\mathbf{K}^a)$ . Equation (10) is the concrete  $4 \times 4$  system

$$\underbrace{\begin{bmatrix} T_{GG} & T_{GI} & T_{GD} & t_G \\ T_{GI} & T_{II} & T_{ID} & t_I \\ T_{GD} & T_{ID} & T_{DD} & t_D \\ t_G & t_I & t_D & r_S \end{bmatrix}}_{\mathbf{T}_S} \underbrace{\begin{bmatrix} \hat{\sigma}_G^2 \\ \hat{\sigma}_I^2 \\ \hat{\sigma}_D^2 \\ \hat{\sigma}_0^2 \end{bmatrix}}_{\mathbf{q}_S} = \underbrace{\begin{bmatrix} q_G \\ q_I \\ q_D \\ r_S \end{bmatrix}}_{\mathbf{q}_S}. \quad (12)$$

Post-projection standardization gives  $t_G = t_I = r_S$ ;  $t_D$ ,  $T_{DD}$ ,  $T_{GD}$ , and  $T_{ID}$  depend on the study environment and design. The last row enforces

$$\hat{\sigma}_G^2 t_G + \hat{\sigma}_I^2 t_I + \hat{\sigma}_D^2 t_D + \hat{\sigma}_0^2 r_S = r_S \quad (13)$$

when the system is full rank. A reported component proportion is therefore  $\hat{\sigma}_a^2 t_a / r_S$ , equivalently its predicted trace contribution divided by the sum of all predicted trace contributions. With  $K$  annotations, the first two rows and columns of Eq. (12) become  $K$ -by- $K$  G and interaction blocks; the N×E and residual rows remain scalar.

For a genetic family  $F \in \{G, I\}$ , define the marginal HE-scale score

$$s_{Sj}^F = \frac{\mathbf{F}_{S:,j}^\top \mathbf{y}_S}{\sqrt{r_S}}. \quad (14)$$

It follows exactly that

$$\mathbf{y}_S^\top \mathbf{K}_{Sk}^F \mathbf{y}_S = \frac{r_S}{M_k} \sum_j A_{jk} (s_{Sj}^F)^2. \quad (15)$$

For one unweighted annotation this fills the first two entries in Eq. (12) as

$$q_G = \frac{r_S}{M} \sum_j (s_{Sj}^G)^2, \quad q_I = \frac{r_S}{M} \sum_j (s_{Sj}^I)^2. \quad (16)$$

The required statistics are marginal cross-products after the common fixed-effect projection. A usual conditional interaction Wald statistic is not interchangeable with Eq. (14) unless its numerator and scale are transformed back to this marginal score.

Let

$$L_{S,jk}^{FH} = \sum_m A_{mk} \left( \frac{\mathbf{F}_{S:,j}^\top \mathbf{H}_{S:,m}}{r_S} \right)^2. \quad (17)$$

Then the genetic block of Eq. (10) is also exact:

$$\text{tr}(\mathbf{K}_{Sk}^F \mathbf{K}_{S\ell}^H) = \frac{r_S^2}{M_k M_\ell} \sum_j A_{jk} L_{S,j\ell}^{FH}. \quad (18)$$

Equations (15) and (18) explain the existing matched-cohort summary implementation. They do *not* justify inserting an independently estimated  $L_R$  into Eq. (18): the finite-cohort trace has both same-person fourth moments and different-person LD moments, which scale differently with cohort size.

## 4 Step 3: transfer the genetic normal-equation block

### 4.1 Population transport assumptions

The reference construction requires the following assumptions.

1. Cohorts  $R$  and  $S$  are independent draws from the same target population, with the same variant/allele and annotation definitions.

2. After applying the declared fixed-effect convention and per-cohort restandardization, the joint second and fourth moments of  $(X, W, E)$  are transportable. Fixed-effect dimension is small relative to cohort size, so the projection's dependence and leverage perturb the exchangeable-individual moments only at lower order. Large leverage differences or population/LD/environment shift violate this assumption.
3. Trait missingness may change  $N_S$  but is ignorable conditional on the modeled variables. Genotypes, environment, and covariates used to form the marginal scores obey the same feature convention as the reference.

These are reference-panel assumptions, not algebraic identities. If the joint G–E distribution differs, the estimator converges to a reference-weighted pseudo-parameter, just as additive SUMMIT with mismatched LD has target  $(\mathbf{A}_R)^{-1} \mathbf{A}_S \mathbf{h}^2$  [3].

## 4.2 Same-person and different-person trace moments

For any genetic kernels  $a = (F, k)$  and  $b = (H, \ell)$ , write

$$\begin{aligned} D_{ab}^C &= \sum_{i=1}^{N_C} K_a^C(i, i) K_b^C(i, i), \\ O_{ab}^C &= \sum_{i \neq i'} K_a^C(i, i') K_b^C(i, i'), \quad T_{ab}^C = D_{ab}^C + O_{ab}^C. \end{aligned} \quad (19)$$

The split is on the individual axis, not an arbitrary orthonormal basis for the residual subspace. Residual rank  $r_C$  controls feature and score normalization, but it is not the number of exchangeable people. Define population moments

$$d_{ab} = \mathbb{E}[K_a(i, i) K_b(i, i)], \quad o_{ab} = \mathbb{E}[K_a(i, i') K_b(i, i')], \quad i \neq i'. \quad (20)$$

Then

$$\mathbb{E}[T_{ab}^S] = N_S d_{ab} + N_S(N_S - 1) o_{ab}. \quad (21)$$

The reference estimators are

$$\hat{d}_{ab} = \frac{\hat{D}_{ab}^R}{N_R}, \quad \hat{o}_{ab} = \frac{\hat{T}_{ab}^R - \hat{D}_{ab}^R}{N_R(N_R - 1)}, \quad (22)$$

and therefore

$$\boxed{\hat{T}_{ab}^{S|R} = \frac{N_S}{N_R} \hat{D}_{ab}^R + \frac{N_S(N_S - 1)}{N_R(N_R - 1)} (\hat{T}_{ab}^R - \hat{D}_{ab}^R).} \quad (23)$$

In matrix form, let  $\mathbf{T}_R^{\text{gen}}$  be the reference genetic trace block and let  $\mathbf{D}_R^{\text{gen}}$  be its same-person part. Both are  $2K$  by  $2K$ , ordered as  $(G_1, \dots, G_K, I_1, \dots, I_K)$ . The block inserted into the trait normal equation is

$$\boxed{\begin{aligned} \hat{\mathbf{T}}_{S|R}^{\text{gen}} &= \frac{N_S}{N_R} \hat{\mathbf{D}}_R^{\text{gen}} \\ &\quad + \frac{N_S(N_S - 1)}{N_R(N_R - 1)} (\hat{\mathbf{T}}_R^{\text{gen}} - \hat{\mathbf{D}}_R^{\text{gen}}) \end{aligned}} \quad (24)$$

Here  $\hat{\mathbf{T}}_R^{\text{gen}}$  is the scaled trace obtained by aggregating the reference directional LD-score panels according to Eq. (18); it is computed once and does not involve a trait. The only trait quantity in

Eq. (24) is  $N_S$ . The trait's GWAS/GWIS scores enter  $\mathbf{q}_S$  through Eq. (15), not the reference trace. This separation is what permits one all-environment reference cohort to be combined with many differently masked phenotypes. When  $N_S = N_R$ , Eq. (23) returns the original reference trace exactly. For additive standardized genotypes,  $D^R$  is close to the familiar  $N_R$ -order baseline. For  $G \times E$ , it contains quantities such as  $\mathbb{E}[X^2 E^2 X'^2 E'^2]$  and is not universal. This is the precise reason that blind squared-rank scaling, or importing the additive leading term, fails. There is no assumption-free way to recover a study's conditional trace from an independent cohort. Equation (23) is the reference-panel estimator under the population-transport assumptions above; without them the target is necessarily reference weighted.

### 4.3 Estimating the fourth-moment term without a second genotype pass

For feature family  $a = (F, k)$  and probe  $v$ , the reference trace pass already forms

$$S_{aiv} = \sum_j F_{R,ij} \sqrt{A_{jk} z_{jv}}, \quad \mathbb{E}_z[S_{aiv}^2] = M_k K_a^R(i, i). \quad (25)$$

For  $B \geq 2$ , an unbiased order-two probe U-statistic is

$$\hat{D}_{ab}^R = \frac{1}{B(B-1)M_k M_\ell} \sum_i \left[ \left( \sum_v S_{aiv}^2 \right) \left( \sum_u S_{biu}^2 \right) - \sum_v S_{aiv}^2 S_{biv}^2 \right]. \quad (26)$$

The subtraction excludes identical probes; different probe indices are independent even when annotations overlap and  $G/G \times E$  sketches share the same underlying probe. An implementation needs an  $N_R \times 2K$  matrix of running square sums and one  $2K \times 2K$  accumulator. Temporary conversion is chunked over probes. Memory is  $16N_R K + O(K^2)$  bytes in float64 and is independent of  $B$ ; at  $N_R = 291,273, K = 1$  it is about 4.4 MiB. No source sketch, block sketch, or cache is written to disk.

## 5 Step 4: complete the normal equation in the trait cohort

### 5.1 $N \times E$ and residual moments

$N \times E$  is study specific and cheap; it should not be borrowed from  $R$ . Let  $\mathbf{Q}_S$  be an orthonormal basis for the fixed effects and  $\mathbf{d}_S = \mathbf{e}_S^2$ . With  $\mathbf{D}_S = \mathbf{P}_S \text{diag}(\mathbf{d}_S) \mathbf{P}_S$ ,

$$\text{tr}(\mathbf{D}_S) = \sum_i d_{Si} - \text{tr}(\mathbf{Q}_S^\top \text{diag}(\mathbf{d}_S) \mathbf{Q}_S), \quad (27)$$

$$\text{tr}(\mathbf{D}_S^2) = \sum_i d_{Si}^2 - 2 \text{tr}(\mathbf{Q}_S^\top \text{diag}(\mathbf{d}_S^2) \mathbf{Q}_S) + \text{tr}\{(\mathbf{Q}_S^\top \text{diag}(\mathbf{d}_S) \mathbf{Q}_S)^2\}. \quad (28)$$

The phenotype supplies

$$q_D = \mathbf{y}_S^\top \mathbf{D}_S \mathbf{y}_S = \|\mathbf{e}_S \odot \mathbf{y}_S\|^2, \quad q_0 = \mathbf{y}_S^\top \mathbf{y}_S = r_S, \quad (29)$$

on SUMMIT's normalized residual scale. Cross-traces with genetic kernels are also exact study summaries. The same pass that computes marginal GWAS/GWIS scores forms

$$d_{Sj}^F = \frac{\mathbf{F}_{S, :, j}^\top \text{diag}(\mathbf{e}_S^2) \mathbf{F}_{S, :, j}}{r_S}, \quad \text{tr}(\mathbf{K}_{S_k}^F \mathbf{D}_S) = \frac{r_S}{M_k} \sum_j A_{jk} d_{Sj}^F, \quad (30)$$

because projected feature columns satisfy  $\mathbf{P}_S \mathbf{F}_{S,:,j} = \mathbf{F}_{S,:,j}$ . This costs no extra genotype decode and avoids transporting an environment-sensitive fourth moment. Because Eq. (5) makes  $\text{tr}(\mathbf{K}_a^S) = r_S$ , genetic-residual cross-traces are exact.

Equations (15), (23), (27)–(30) fill every entry of the study normal equations. Solving them estimates the four classes ( $G, G \times E, N \times E$ , residual) using genotype/environment moments from all eligible reference individuals and trait-specific marginal scores from the study cohort.

## 6 Step 5: trait summaries, solution, and uncertainty

### 6.1 Sufficient summary data and fitting algorithm

The reusable reference object contains the ordered variant/allele and annotation definitions,  $N_R, r_R$ , the randomized genetic trace matrix  $\hat{\mathbf{T}}_R$ , and the compact same-individual matrix  $\hat{\mathbf{D}}_R$ . These quantities are phenotype independent and are computed once in every individual with an observed environment and reference covariates. For each trait, the required study object is

$$\mathcal{S}_y = \{N_S, r_S, \mathbf{s}_S^G, \mathbf{s}_S^I, q_D, q_0, \text{tr}(\mathbf{D}_S), \text{tr}(\mathbf{D}_S^2), \text{tr}(\mathbf{K}_{S_k}^G \mathbf{D}_S), \text{tr}(\mathbf{K}_{S_k}^I \mathbf{D}_S)\}_{k=1}^K. \quad (31)$$

The two score vectors are marginal GWAS/GWIS cross-products. The remaining terms are scalars or  $O(K)$  design summaries. Equation (30) lets the genetic- $N \times E$  terms be accumulated in the same streamed genotype pass that produces GWAS/GWIS, with no randomized trace calculation and no second decode. Once  $\mathcal{S}_y$  is written, fitting uses no individual-level phenotype, genotype, or environment. Thus trait missingness affects the cheap trait summary but does not force reconstruction of the all- $E$   $G \times E$  reference.

For each phenotype, the complete fitting algorithm is therefore:

1. retain individuals with phenotype, environment, and covariates; standardize  $E$ , build  $\mathbf{P}_S$ , and normalize  $\mathbf{y}_S$  as in Eq. (3);
2. stream the genotype once to form  $\mathbf{s}_S^G, \mathbf{s}_S^I$ , and the exact genetic- $N \times E$  traces in Eq. (30);
3. compute  $q_D, q_0, t_D, T_{DD}$  from  $\mathbf{e}_S, \mathbf{y}_S$ , and the fixed-effect basis;
4. transfer only the reference genetic block with Eq. (24), assemble Eq. (12), and solve for  $\hat{\boldsymbol{\theta}}$ ;
5. report  $\hat{\sigma}_a^2 t_a / r_S$  and obtain standard errors from variant-block delete-one fits.

### 6.2 Missing genotypes and heterogeneous per-variant samples

The current production contract uses one analysis cohort for every retained variant. In the trait-score pass, a missing genotype at variant  $j$  is replaced by the observed mean of that variant within the retained trait cohort. After centering, the imputed entry is zero. The additive and interaction columns are then projected and restandardized exactly as in Eq. (5). Consequently all output rows carry the same  $N_S$  and  $r_S$  even if a small number of raw calls were missing. This is a coherent mean-imputation approximation under ignorable, low-rate missingness; it is not a correction for genotype missingness associated with phenotype or environment.

The bounded native reference builder is stricter: it requires the selected BED to be missing-free and aborts if it encounters a missing call. Thus a production reference should use dosage-imputed or otherwise explicitly imputed genotypes, as in the EUR 300k analysis, or a missing-free hard-call set after quality control. Reference and study features are always centered, projected, and restandardized within their own retained cohorts.

Variant-specific summary sample sizes  $N_j$  are not currently supported. The score reader verifies that every GWAS and GWIS row has the phenotype’s common  $N_S$  and  $r_S$ . This is deliberate: if variants  $j$  and  $m$  are observed in different individuals, their contribution to  $\text{tr}(\mathbf{K}_a \mathbf{K}_b)$  depends on the pairwise overlap and on the two mask-specific projectors, not only on  $N_j$  and  $N_m$ . Standard summary statistics containing only per-variant  $N_j$  do not identify those terms. Simply substituting  $N_j$ , an average  $N$ , or  $\sqrt{N_j N_m}$  into Eqs. (15)–(18) is therefore not exact. A future mask-aware estimator would need at least overlap summaries (or a model for the missingness masks) and consistently defined mask-specific GWAS, GWIS, and LD moments. Until then, variants with appreciable or differential missingness should be imputed and quality controlled before constructing the SUMMIT summaries.

### 6.3 Uncertainty and identifiability

There are three uncertainty sources: trait-score sampling, finite reference sampling, and randomized-probe error. A SNP-block jackknife deletes the corresponding GWAS/GWIS score rows and reference LD-score rows. The default block-local approximation assumes cross-block LD is negligible and reuses the full-reference  $\hat{D}^R$  in each replicate. This omits a change of order the deleted SNP fraction in the diagonal term, so it is an explicit approximation whose stability should be checked when blocks are few, large, or annotation-imbalanced. An exact deletion would require block-specific cross-sketch moments and is intentionally not the production default.

Increasing  $B$  reduces only probe error. It cannot repair reference mismatch or an unidentified model. In particular, if  $E^2$  is constant after projection,  $\mathbf{D}_S$  is proportional to  $\mathbf{P}_S$  and  $\mathbf{N} \times \mathbf{E}$  is aliased with residual noise. Collinear annotation kernels cause analogous rank failure.

## 7 Validation

Three independent checks were used.

1. A dense oracle forms all individual-level kernels and verifies that the matched summary equations reproduce every GENIE matrix, right-hand side, and trace entry to floating-point tolerance (maximum tested discrepancy below  $2 \times 10^{-13}$ ).
2. Setting  $N_S = N_R$  in the implemented transfer reproduces the complete reference genetic block exactly, irrespective of how  $T_R$  is split into  $D_R$  and  $O_R$ .
3. In a separate continuous- $E$  Monte Carlo experiment, an independent  $N_R = 2500$  reference and 60 independent  $N_S = 350$  studies used  $M = 60$  correlated Gaussian markers (AR(1) correlation 0.35), with intercept and  $E$  projected out. Median relative errors of Eq. (23) were 2.17% (G–G), 3.14% (G–G×E), and 2.23% (G×E–G×E). Blind squared-rank scaling gave 9.27%, 86.46%, and 23.51%, respectively. Genetic- $\mathbf{N} \times \mathbf{E}$  cross-traces were computed exactly in each study rather than transported. These are finite-reference errors, not claimed universal operating characteristics.

An end-to-end integration test also constructs one reference, removes a trait value to create a different study cohort, writes the study GWAS/GWIS moments, and fits the transferred equations.

## 8 Covariates and restandardization in released GENIE code

The equations in the GENIE paper are coherent: with  $\mathbf{V} = \mathbf{I} - \mathbf{C}(\mathbf{C}^\top \mathbf{C})^{-1} \mathbf{C}^\top$ , Supplemental Note S1 uses  $\text{tr}(\mathbf{K}_k \mathbf{V} \mathbf{K}_\ell \mathbf{V})$ ,  $\text{tr}(\mathbf{V} \mathbf{K}_k)$ ,  $\mathbf{y}^\top \mathbf{V} \mathbf{K}_k \mathbf{V} \mathbf{y}$ , and residual dimension  $N - C$  [2]. The concerns

below apply to the released code at commit 22bee9c, not to that formal derivation.

**Continuous-environment scale and missing values.** The released GENIE reader copies finite environmental values directly into the design and interaction terms; it does not center or standardize a continuous environment internally [5]. A GENIE analysis must therefore supply an environment already expressed in the intended units, normally mean zero and unit standard deviation. SUMMIT instead applies Eq. (2) internally and records the transformation. This distinction is substantive because multiplying  $E$  by  $c$  multiplies the  $G \times E$  feature by  $c$  and the  $N \times E$  kernel by  $c^2$ , thereby changing their coefficient scales. The same GENIE reader masks the literal string NA, but its -9 branch skips assignment without clearing the analysis mask [5]; unless the caller removes those rows, the zero-initialized value is treated as an observed environment.

**Ordinary inverse and column count.** The implementation computes  $Q = WtW.inverse()$  and sets the residual dimension to  $N_{indv\_mask} - N_{cov}$  [4]. This is correct only if the retained covariate columns are exactly full rank and numerically well conditioned. Environment, intercept, duplicated indicators, or a constant covariate can make that false. A rank-revealing QR/SVD and  $N - \text{rank}(C)$  are required for the projector in Eq. (1).

**Scaling before the analysis mask.** The genotype code computes allele means over  $N_{indv}$  and applies the HWE scale  $\{2p(1-p)\}^{-1/2}$  before the phenotype/environment mask and fixed-effect projection [6]. Consequently a retained, projected column generally does not have squared norm  $N_{mask}$  or  $N_{mask} - \text{rank}(C)$ . This is especially visible when trait missingness creates a much smaller analysis sample.

**The important distinction: two coherent estimands and one incoherent hybrid.** There are two valid choices.

1. *Raw projected kernel:* use  $U = PX$  and  $W = P(E \odot X)$  without post-projection scaling. This defines a valid model, but every trace, including  $\text{tr}(K)$ , must be computed from the realized columns and the random effect prior is tied to the pre-projection genotype scale.
2. *Restandardized projected kernel:* rescale every projected column as in Eq. (5). Then all genetic kernel traces equal  $r$  and summary moments have a stable, cohort-comparable per-residual-SD definition. This is SUMMIT’s default.

The released GENIE implementation initializes each additive genetic trace to  $N_{indv\_mask}$ , while interaction/noise traces are estimated from the random projections, and then applies a projection subtraction [7]. That mixes the raw-kernel Gram and phenotype moments with a unit-norm additive trace assumption that the preceding HWE scaling/masking does not guarantee. The problem is not simply “GENIE fails to restandardize”: raw projected kernels are legitimate. The problem is using their realized Gram/RHS while hard-coding a trace from the alternative standardized-kernel convention.

**Phenotype scaling.** The default code standardizes the phenotype, projects covariates, and then standardizes the projected phenotype again using denominator  $N_{mask} - 1$  [8], while the residual kernel dimension is declared  $N_{mask} - N_{cov}$ . Because the code retains the realized  $\mathbf{y}^\top \mathbf{y}$  in the normal equations, phenotype rescaling alone is not a bias. GENIE’s reported results under this default are residual-phenotype (conditional) proportions, so the closest SUMMIT comparison is SUMMIT’s

primary conditional proportion, not the optional value multiplied by the post-covariate variance fraction. The remaining  $N - 1$  versus  $r$  convention must still be propagated consistently through kernel traces; it is the combination with hard-coded trace conventions, rather than residualizing the phenotype itself, that is problematic.

These issues can be small when covariates are full rank, missingness is mild, and HWE scaling nearly matches the retained sample variance. They are not algebraically controlled in the released implementation. SUMMIT instead uses a rank-revealing orthonormal basis, records the numerical rank, applies the trait/reference mask before feature construction, and either restandardizes projected columns or computes their actual traces consistently.

## 9 Computational implications

The new population statistic does not change the dominant complexity. With  $B$  probes,  $K$  annotations,  $M$  variants, and streamed genotype blocks, the G/G×E reference remains linear in  $N_R$ ,  $M$ ,  $B$ , and  $K$  for  $K = 1$  (and in the number of requested directional annotation products generally). The added Eq. (26) work is  $O(N_RBK^2)$  but has a very small constant and no genotype decode or large matrix multiply; its persistent memory is  $O(N_RK)$ .

On tabla, a sealed full  $N_R = 291,273$ ,  $M = 454,207$ ,  $B = 32$  EUR age reference completed its numerical work in 1,503.6 seconds; end-to-end measured elapsed time was 1,534.9 seconds (25.6 minutes), with 4.04 GiB peak RSS. Its feature, source, panel-preparation, and target phases took 560.0, 368.6, 0.20, and 471.4 seconds. The source tree, loaded native binary, and single OpenBLAS 0.3.34 runtime are hash-bound in the published reference manifest. Relative to the preceding internally threaded diagnostic, total elapsed time fell 20.9%, target time fell 51.9%, and repaired output columns fell from 384 to 3 without increasing memory.

A cohort-matched additive-only  $B = 32$  run used the same  $N = 291,273$ ,  $M = 454,207$ , covariates, probe distribution and seed, 32-core affinity, and 16-GiB sketch budget. It completed in 568.4 seconds end to end (565.4 seconds reported numerical runtime) with 1.39 GiB peak RSS. The corresponding G×E ratios are therefore 2.70-fold for elapsed time and 2.90-fold for peak RSS. The extra work is intrinsic to constructing the interaction features and estimating the four directional G/G×E kernel products instead of the one additive product; it is not disk-cache I/O. The comparison produced only a 4.97-MB additive LD-score file in addition to its small log and receipt.

A direct  $B/32$  extrapolation is too pessimistic because the  $B = 32$  products are skinny. In a dense-oracle benchmark with  $N = 100,000$  and  $M = 1,000$ , increasing  $B$  from 32 to 512 increased native source time from 0.319 to 1.267 seconds (3.97-fold) and the relevant  $2B$  target time from 0.458 to 2.396 seconds (5.23-fold), rather than 16-fold. A subsequent direct  $B = 1024$  run measured 2.026 seconds for the source product, 4.485 seconds for the  $2B$  target, and 8.453 seconds for the  $4B$  target. Its maximum target error against the independent dense float64 oracle was  $2.538 \times 10^{-7}$ , with no repaired columns. Scaling the sealed full-cohort  $B = 32$  phases by these measured  $B = 1024/B = 32$  ratios gives approximately 2.1 hours. A conservative current planning range is 2.2–2.8 hours to allow for repair and system-load variation. The  $B = 1024$  component times are measured, but the full 454,207-variant time is still a scaling estimate and should be replaced by an end-to-end measurement before a large production campaign.

Probe tiling and block-width adaptation bound memory. For one annotation, the float32  $[XZ, WZ]$  source sketch costs  $8N_RB$  bytes. The native opaque  $[S, E \odot S]$  target panel costs  $32N_RB$  bytes, while the protected non-decoded target operand costs at most another  $16N_RB$  bytes *inside* the configured native workspace. At  $N_R = 291,273$  and  $B = 1024$ , these terms are about 2.22, 8.89, and 4.44 GiB. The first two allocations plus the bounded 16-GiB native workspace (whose variant

block shrinks as the protected operand grows) imply an expected peak of roughly 25–30 GiB, not 32 times the  $B = 32$  RSS. The new population-transfer statistic itself adds only about 4.4 MiB for  $K = 1$ .

The numerical failures are not caused by extreme Rademacher probes: their entries are exactly  $\pm 1$ . Nor is float32 the arithmetic root cause. The decoder, feature moments, projection, source/target GEMMs, checksums, and repairs use float64; float32 controls only retained sketch buffers. Switching those buffers to float64 doubles their memory and reduces accumulation roundoff, but cannot prevent a vendor GEMM from sparsely changing a double input or output. Dropping extreme LD-score values would be unsafe here: an extreme Monte Carlo realization and a corrupt output are different events, and clipping squared traces creates downward bias.

Before the environment repair, *tabla* loaded two BLAS runtimes in one process: NumPy’s pip wheel bundled OpenBLAS 0.3.30/Haswell while the native extension used conda OpenBLAS 0.3.34/Zen. The environment has now been changed to conda NumPy 2.3.5, so NumPy and the extension both resolve to the same OpenBLAS 0.3.34 Zen library. This removes a concrete source of process-wide thread and runtime interference, but it did not remove the numerical fault. The full-cohort  $B = 32$  diagnostic triggered 384 repaired output columns. More decisively, a production-shaped dense test with  $N = 291,273$ ,  $M = 1,207$ , and  $B = 32$  intermittently observed a maximum target error of 58.15 from the internally threaded vendor path, despite using binary64 arithmetic. Repeating the same test could also return only ordinary accumulation-order error, proving that the failure is intermittent and shape/thread dependent.

SUMMIT therefore no longer invokes an internally threaded OpenBLAS GEMM for these products. It fixes OpenBLAS at one thread per call and partitions disjoint output rows across SUMMIT-owned OpenMP workers. This was faster, not slower, in the production-shaped no-fault comparison (1.09 versus 1.33 seconds for the  $2B$  target and 1.79 versus 2.28 seconds for  $4B$ ). Across 20 repeated  $2B/4B$  calls, the selective ABFT layer repaired 257 output columns and every final output remained within  $2.283 \times 10^{-6}$  of the independent dense float64 oracle. Thus the vendor/concurrency problem is still observable, but the returned result is protected without duplicating the full genotype operand. The ABFT path checks wide GEMMs, repairs only discrepant output columns with a non-vendor cache-tiled kernel, and falls back to a deterministic fresh decode only if the reconstructable genotype input changes.

A controlled microkernel experiment held the OpenBLAS 0.3.34 library, input, thread count, and 20-call stress sequence fixed, but forced dynamic dispatch from Zen to Haswell. Repaired columns fell from 257 to 1 (99.6%), while source,  $2B$  target, and  $4B$  target time increased by 29%, 18%, and 6%, respectively. This strongly localizes the residual failure to the Zen dispatch or microkernel path rather than genotype decoding, probe values, or float32 storage. Haswell is not yet safe enough to remove integrity checks, and the sealed full-cohort Zen run required only three repairs, so the faster Zen path with selective ABFT remains the production default. A truly non-threaded OpenBLAS build or an independent BLAS implementation is the next clean test of the remaining vendor-runtime hypothesis.

## 10 Real-data validation status

The implemented unequal-cohort path has been exercised with height and age: the reference included every eligible individual with age and covariates, whereas the GWAS/GWIS and study-specific  $N \times E$  moments used all available height observations. The fit was full rank and numerically solved its normal equation. Its additive and interaction estimates were close to those from the earlier common-complete-case analysis, which is supportive of reference transport within this ancestry

group. It is not a controlled equivalence test, because both the reference size and the trait-score sample changed.

A decisive empirical validation should hold the trait cohort fixed and compare (i) an in-sample genetic Gram matrix with (ii) the population-transferred reference matrix, using identical scores, feature definitions, variants, blocks, and probe precision. Across many traits one should report component differences relative to their joint jackknife uncertainty, normal-equation conditioning, and sensitivity to reference size and probe count. The present simulation/oracle tests establish the algebra and implementation; the single-trait result establishes end-to-end feasibility but does not replace that matched validation design.

## 11 Practical contract

The implemented external path is explicit. A reference generated with at least two probes records the compact population trace matrix in its manifest. Trait scoring with `--gxe-population-reference` permits a different retained sample, recomputes study feature standardization, and writes study  $N$ ,  $r$ ,  $q_D$ ,  $q_0$ ,  $\text{tr}(D)$ ,  $\text{tr}(D^2)$ , and exact genetic- $D$  cross-traces (plus delete-block versions when requested). The fit requires the exact reference-manifest hash, identical ordered variants/alleles and annotations, the same named environment/covariate convention, and standardized kernels. It fails closed for GENIE/raw kernels because their genetic traces are not fixed by  $r$ . Existing matched-cohort behavior remains the default.

## References

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